Ex.No : 05

Date : 30.7.25

Experiment on packet capture tool : Wireshark

Aim:

To understand the feature of wireshark as a Packet capture tool and understand encapsulation of information at various layers of a protocol stack.

Packet Sniffer

* Captures packets sent and received by the computer
* Displays protocol information of captured packets
* Works passively
* Receives copies of network packets for analysis.

Packet Sniffer Tools :

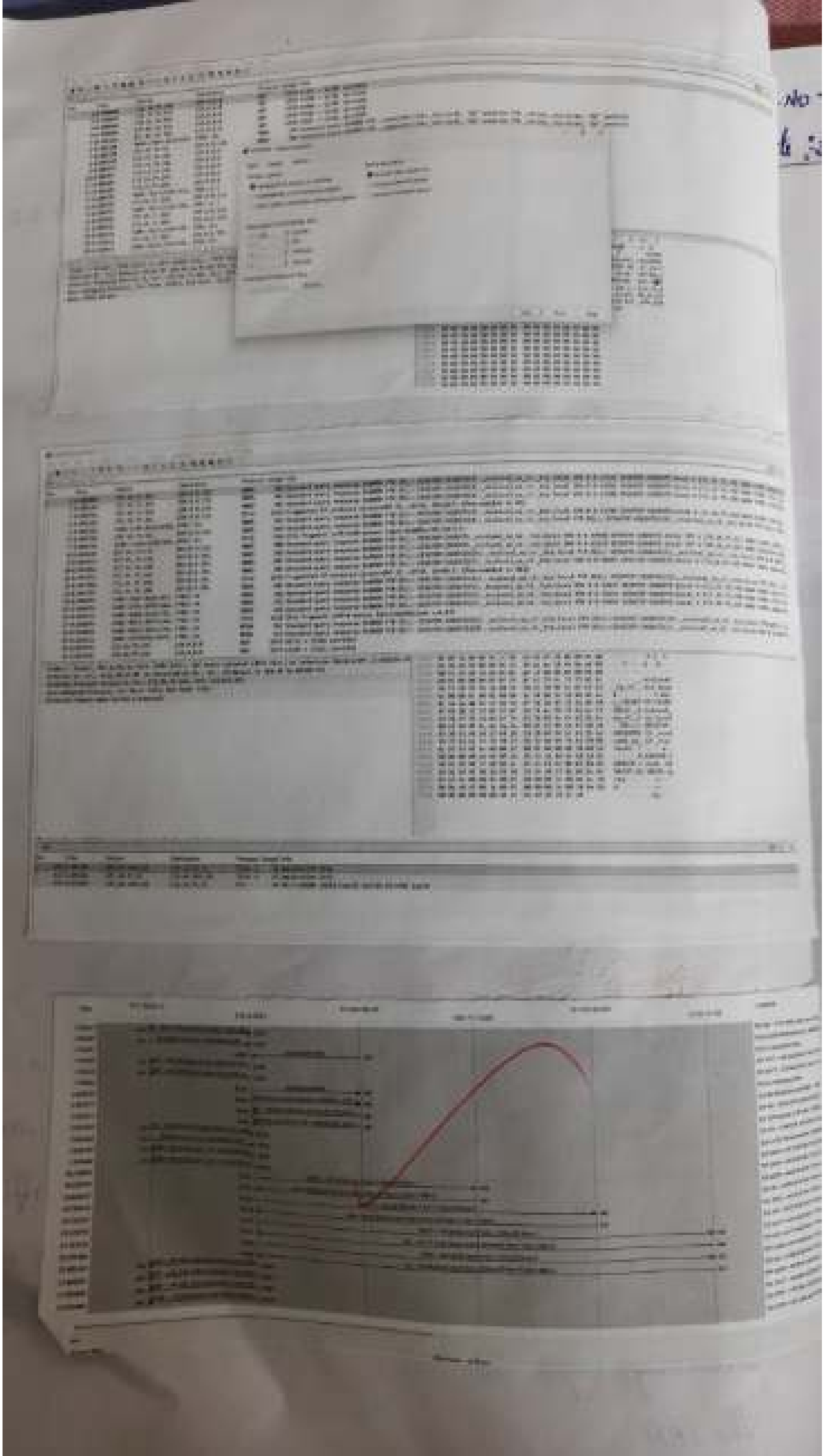
* Tcpdump - Command line packet capture tool
* Wireshark - Graphical packet analyzer

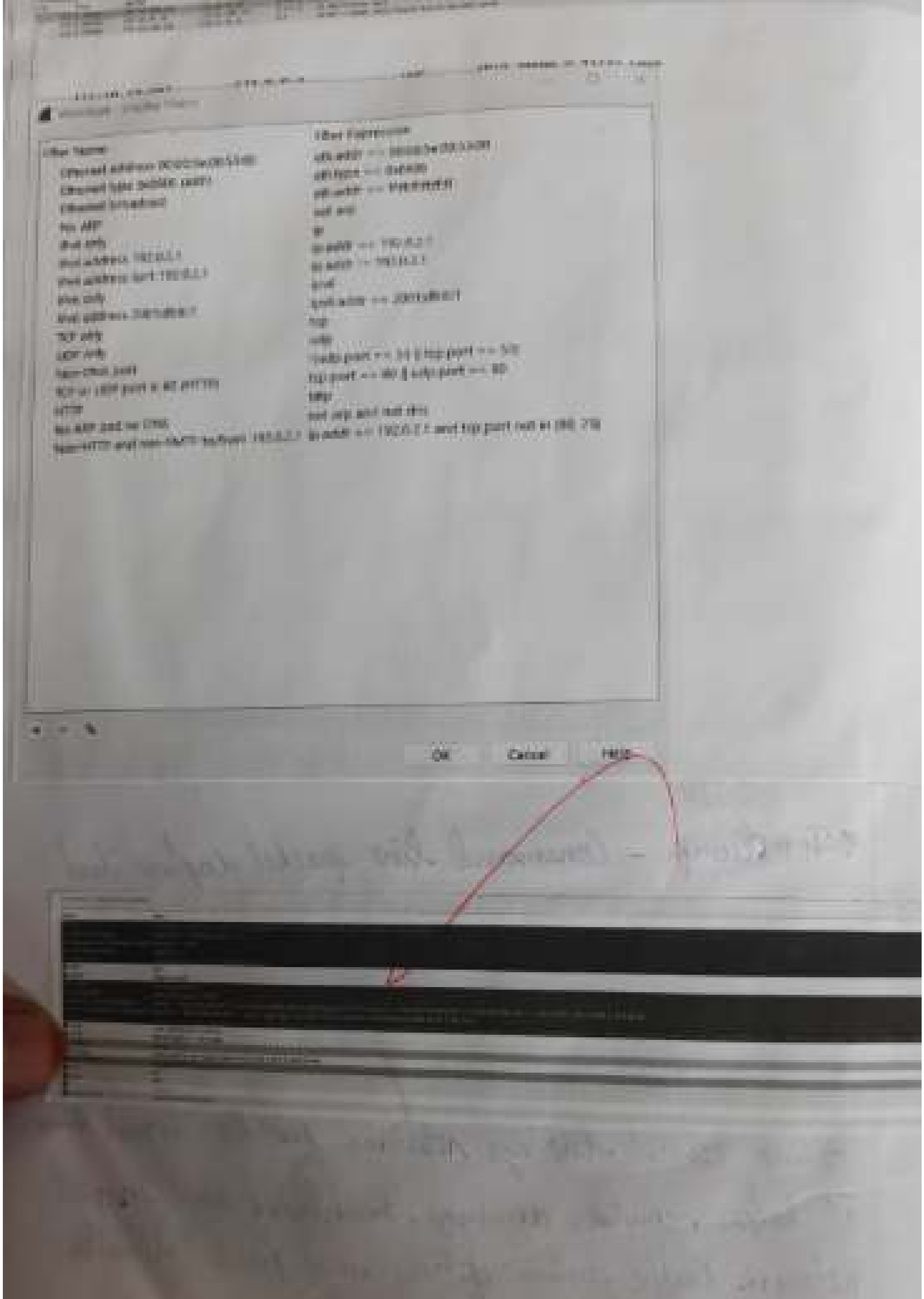
Description :

Wireshark is a network protocol analyzer. It captures and displays network packets in real time. It helps monitor, analyze, troubleshoot and inspect network traffic using filters and packet details.

Uses of Networks :-

* Captures network traffic
* Analyzes protocols
* Applies display filters
* Monitors network usage





Applications :

* network troubleshooting
* Security analysis
* Protocol debugging
* Learning network protocols

Capturing packets :

1. Open Wireshark
2. Select the network interface.
3. Click start Capture
4. Observe Packet in real time
5. Click Stop and Save the capture

Wireshark Panes

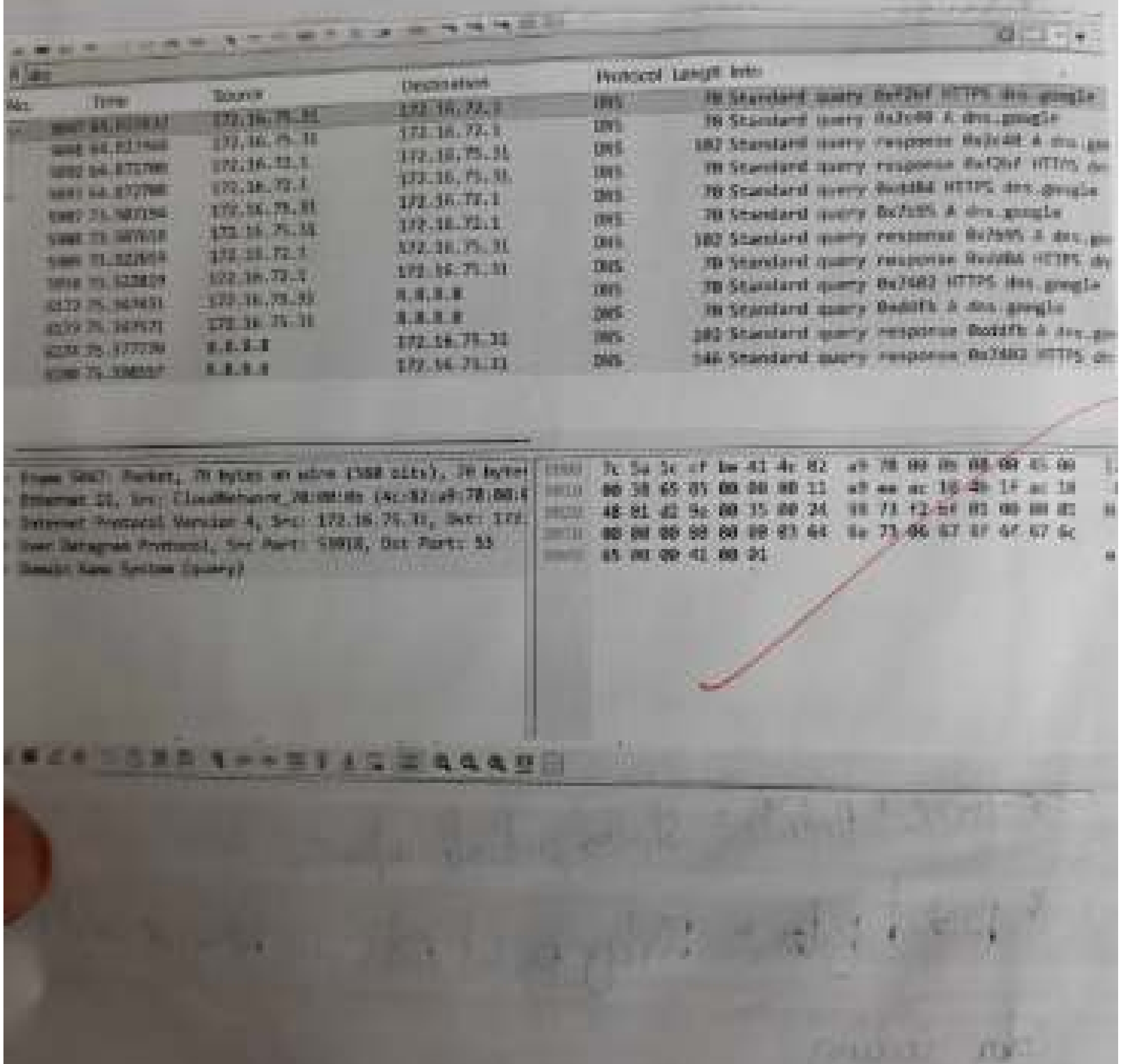
* Packet List :- Display captured packets
* Packet Details :- Shows protocols information
* Packet Bytes :- Display packet data in hexadecimal form

Color Coding :

* Purple :- TCP
* Blue :- UDP
* Black :- Error packets.

Filtering packets :

* Enter a filter in the filter bar
* Right Click to apply filters or follow the TCP Stream


No. Time Source Destination Protocol Length Info
Frame 5867: Packet, 70 bytes on wire (560 bits), 70 bytes
Ethernet II, Src: CloudNetwork_78:00:0b (a2:81:e9:78:00:0
Internet Protocol Version 4, Src: 172.16.75.31, Dst: 172.
User Datagram Protocol, Src Port: 59918, Dst Port: 53
Domain Name System (query)

Flow Graph:

Displays the communication flow between devices, making packing exchange easier to understand.

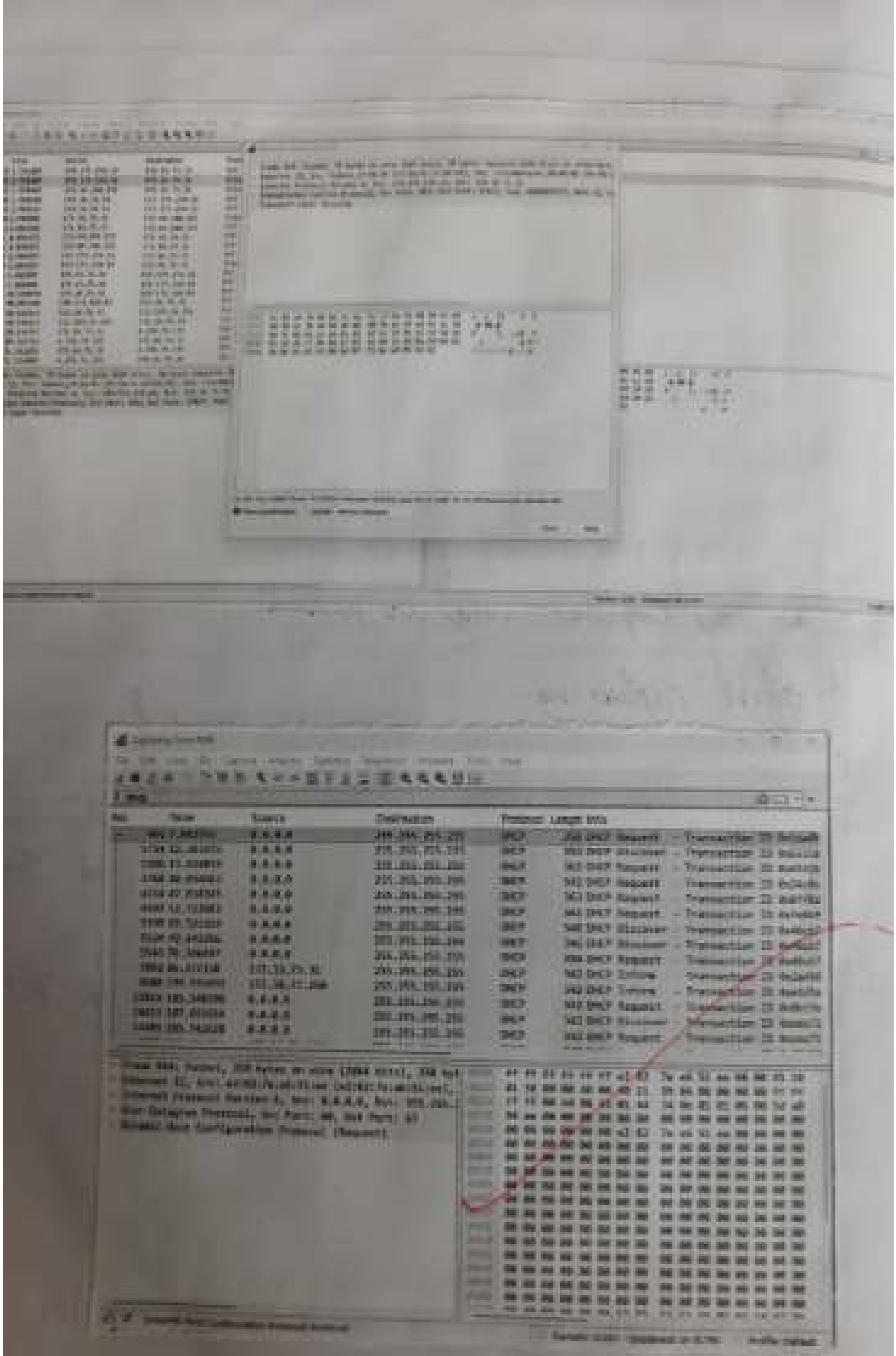
Capture 100 Packets

1. Open Wireshark
2. Select the network interface
3. Set the capture limit into 100 packets
4. Start capturing
5. Stop and save the captured Packets

Analysing Packets using Wireshark Tool.

1. Create a filter to Displays only TCP/UDP Packets.

1. Open the Capture file
2. Apply the TCP || udp filter
3. Inspect and View the flow graph.



2. Create a file to display only ARP Packets

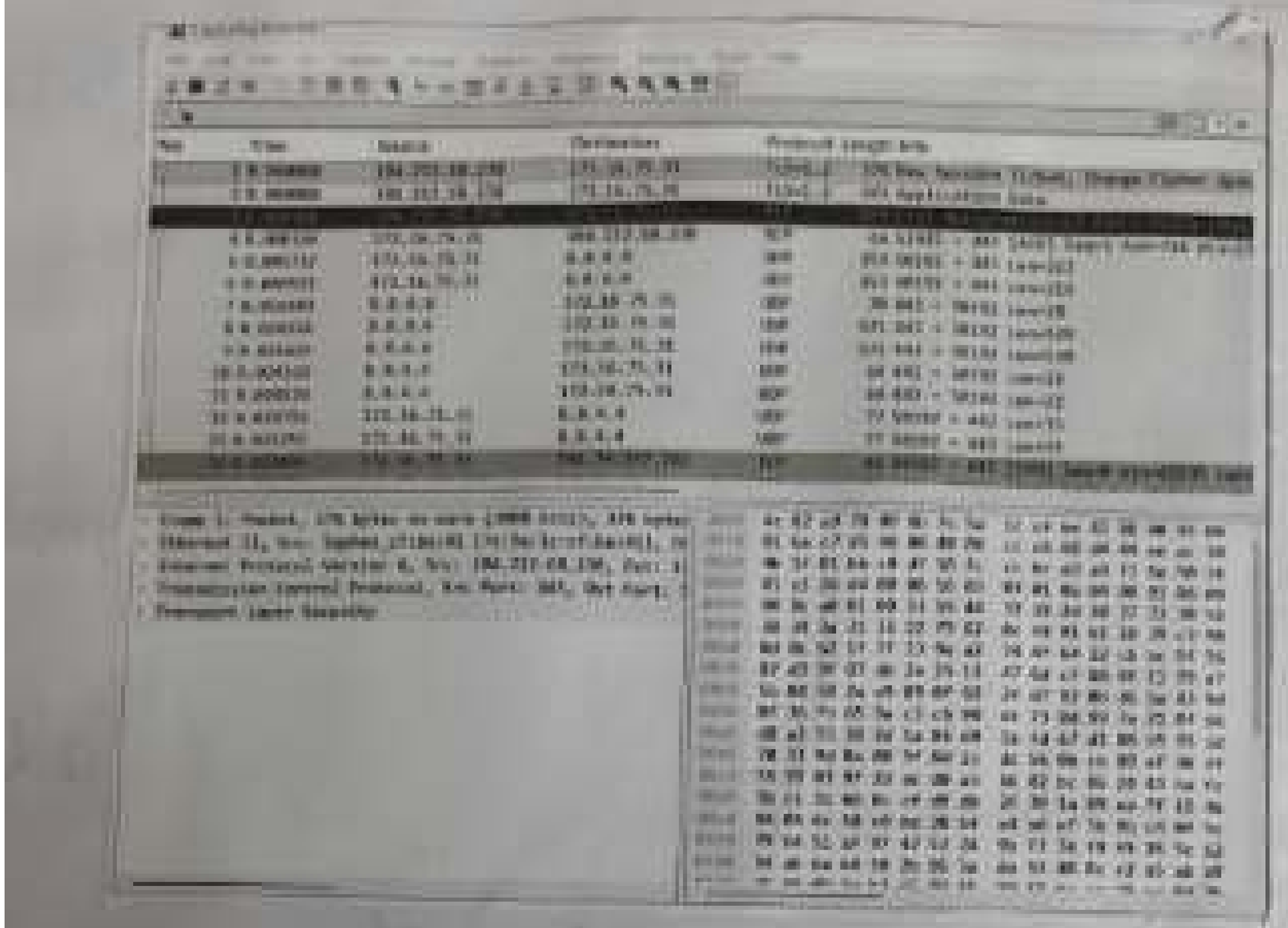
1. Open the capture file
2. Apply the arp filter
3. Inspect and view the flow graph

3. Create a filter to display only DNS packets

1. Open the capture file
2. Apply the DNS filter
3. Inspect and view the flow graph

4. Create a filter to display only IP/ICMP Packets

1. Open the capture file
2. Apply the IP or icmp filter
3. Inspect and view the flow graph and save capture



Time	Source	Destination	Protocol	Length	Info
		Broadcast	ARP	42	Who has 172.16.75.129? Tell 172.16.75.1
		Broadcast	ARP	42	Who has 172.16.77.183? Tell 172.16.75.6
		Broadcast	ARP	42	Who has 172.16.77.183? Tell 172.16.75.6
		Broadcast	ARP	42	Who has 172.16.78.57? Tell 172.16.75.61
		Broadcast	ARP	42	Who has 172.16.75.204? Tell 172.16.75.4
		Broadcast	ARP	42	Who has 172.16.77.91? Tell 172.16.75.5

5 Create a filter to display only DHCP packet

1. Open the capture file
2. Apply the dhcp filter
3. Inspect and view the flow graph and save the capture. -

Result :-

Thus the feature of wireshark and the implement of different filter was Succesfully Completed.